Compiled Problems – Senior Division

1. **01-02 C2 Bit String Flicking**

   Solve for X:

   X= (RCIRC – 2 11000) XOR (LSHIFT- 1 01100)

2. **01-02 C2 Bit String Flicking**

   How many values of X satisfy the following equation?

   (RCIRC – 3 (01010 OR 00101 AND X)) = 11001

3. **02-03 C2 Bit String Flicking**

   Simplify the following expression:

   ((RCIRC–2 01101 OR LSHIFT–1 01011) AND (NOT (LCIRC–3 (RSHIFT–2 01110))))

4. **02-03 C2 Bit String Flicking**

   Solve for X (X is a 5-bit string)

   (LCIRC–3 (RSHIFT–2 (NOT 01001 AND X))) = (RCIRC–1 (LSHIFT–1 01000))

   .

5. **03-04 C2 Bit String Flicking**

   Evaluate the following expression.

   ((NOT (RCIRC-2 (LSHIFT-1 10110))) OR (LCIRC-3 01110) AND (NOT 01011))

6. **03-04 C2 Bit String Flicking**

   Find all values of X (a 5-bit string) that satisfy the following expression.

   ((10110 AND LCIRC-2 X) OR RCIRC-2 X) = 00010

7. **04-05 C2 Bit String Flicking**

   Simplify:

   (RCIRC-2 (LSHIFT-1 (LCIRC-3 (RSHIFT-2 01101)))) OR (NOT (10110 AND 01010))

8. **04-05 C2 Bit String Flicking**

   Solve for X (a 5-bit string):

   (NOT (RCIRC-2 10110)) OR (RCIRC-2 X) = (LCIRC-3 11101 AND 11110)

9. **05-06 C2 Bit String Flicking**

   Define a new operation SWAP-S,L,P as follows: Exchange the bits starting in position S with a length of L with the bits starting in position P and with the same length. Evaluate:

   (RCIRC-3 (SWAP-2,2,4 (LSHIFT-2 0011011)))

10. **05-06 C2 Bit String Flicking**

    How many 5-bit strings X solve the following equation:

    (RCIRC-3 (LSHIFT –1 X OR 01101)) = 11101

11. **06-07 C2 Bit String Flicking**

    The two arguments in the expression below are hexadecimal representations of bit strings that are 12-bits long.
    Evaluate the expression and express your answer as a 3-digit hexadecimal string.

    A26 XOR 7B5

12. **06-07 C2 Bit String Flicking**

    List all the values of X (a 5-bit string) that make the expression TRUE.

    ((LCIRC-3 (X AND 10010)) OR (RSHIFT-2 (X AND 01101)) = 10011

13. **07-08 C2 Bit String Flicking**

    Evaluate the following expression:

    (RCIRC-3 01101 OR (LSHIFT-2 (RCIRC-1 01110) AND 10111))

14. **07-08 C2 Bit String Flicking**

    List all the values of X (a 5-bit string) for which the following expression has a value of 00101.

    ((LCIRC-2 (RSHIFT-1 NOT X)) AND (RCIRC-1 (LCIRC-3 01101)))

15. **08-09 C2 Bit String Flicking**

Evaluate the following hexadecimal expression and express the answer in octal:

DEB XOR ACE AND BAD

16. **08-09 C2 Bit String Flicking**

    List all of the values of X (a 5-bit string) that satisfy :

    (RCIRC-2 (RSHIFT-1 X)) = (RSHIFT-1 10101) OR (LCIRC-2 01010)

17. **09-10 C2 Bit String Flicking**

    Evaluate the following expression:

    (LCIRC-2 (RSHIFT-2 01101)) OR (NOT 10110 AND RCIRC-3 10110) XOR
    (RSHIFT-2 (LCIRC-3 NOT 01100))

18. **09-10 C2 Bit String Flicking**

    How many unique values of X (5 bits long) solve the following equation:

    (RCIRC-3 (LSHIFT-2 (NOT X))) AND (RSHIFT-1(LCIRC-2 X)) = 00001

19. **10-11 C2 Bit String Flicking**

    Evaluate the following expression:

    (LCIRC-2 11011) OR (NOT 11010) AND (RSHIFT-1 (LCIRC-2 01100 AND 10101))

20. **10-11 C2 Bit String Flicking**

    List all possible values of X (5 bits long) that solve the following equation:

    (LCIRC-2 (RSHIFT-1 (NOT (RCIRC-3 X)))) = 10100

21. **11-12 C2 Bit String Flicking**

    Evaluate the following expression:

    (LCIRC−2 01110) XOR (RSHIFT−1 01011)

22. **11-12 C2 Bit String Flicking**

Solve the following for X (5-bit string) that satisfies the following:

(RSHIFT−1 (LCIRC−2 X)) OR (NOT 10111) = 01011

23. **11-12 C3 Bit String Flicking**

    Solve the following for X (5-bit string) that satisfies the equation:

    (RSHIFT-1 (LCIRC-2 X)) =
    (RCIRC-2 01101) OR (LSHIFT-2 10101) AND (NOT 11001)

24. **12-13 C2 Bit String Flicking**

    Evaluate the following expression:

    (LCIRC-2 (RSHIFT-1 (NOT 01110)))

25. **12-13 C2 Bit String Flicking**

    Evaluate the following expression:

    ((RCIRC-2 (01010 XOR 10010)) OR (LSHIFT-1 (NOT 01110)))

26. **12-13 C3 Bit String Flicking**

    Solve the following for X (5-bit string) that satisfies the equation:

    (LCIRC-2 01011) OR X = RSHIFT-1 11011

27. **13-14 C2 Bit String Flicking**

    Evaluate the following expression:

    (RSHIFT-2 (LCIRC-1 01011)) OR (NOT (LSHIFT-1 (RCIRC-3 10110))))

28. **13-14 C2 Bit String Flicking**

    Solve for X (5-bits) in the following equation:

    (RCIRC-2 (LSHIFT-1 X)) = 10111

29. **13-14 C3 Bit String Flicking**

Solve the following for X (5-bit string) that satisfies the equation:

(LSHIFT-2 (RCIRC-1 01110)) OR (LCIRC-2 (RSHIFT-1 X)) AND
(NOT (RCIRC-1 (LSHIFT-2 01010))) = (LCIRC-2 (NOT 10000))

30. **14-15 C2 Bit String Flicking**

Evaluate the following expression:

(LSHIFT-2 (RSHIFT-1 01101)) OR ((LCIRC-2 (RSHIFT-1 01110)) AND (NOT 01100))

31. **14-15 C2 Bit String Flicking**

Solve for X (5-bits) in the following equation. How many unique solutions are there?

(RCIRC-2 (LSHIFT-1 (NOT X))) = 00101

32. **14-15 C3 Bit String Flicking**

Solve the following for X (5-bit string) that satisfies the equation:

NOT ((LCIRC-2 01011) AND (RSHIFT-1 X)) = (RCIRC-2 01110 OR 10110 AND NOT X)

33. **15-16 C2 Bit String Flicking**

Evaluate the following expression:

(LCIRC-2 10011) OR (RSHIFT-2 01110) AND (RCIRC-2 (LSHIFT-1 01010))

34. **15-16 C2 Bit String Flicking**

Solve for X (5-bits) in the following equation.

(LCIRC-2 10001) OR X = (RSHIFT-1 01110)

35. **15-16 C3 Bit String Flicking**

Solve the following for X (5-bit string) that satisfies the equation:

((RSHIFT-1 (LCIRC-2 (LSHIFT-1 X))) OR (NOT 01110 AND (RCIRC-2 01101))) =
(LCIRC-1 (LSHIFT-2 00100))